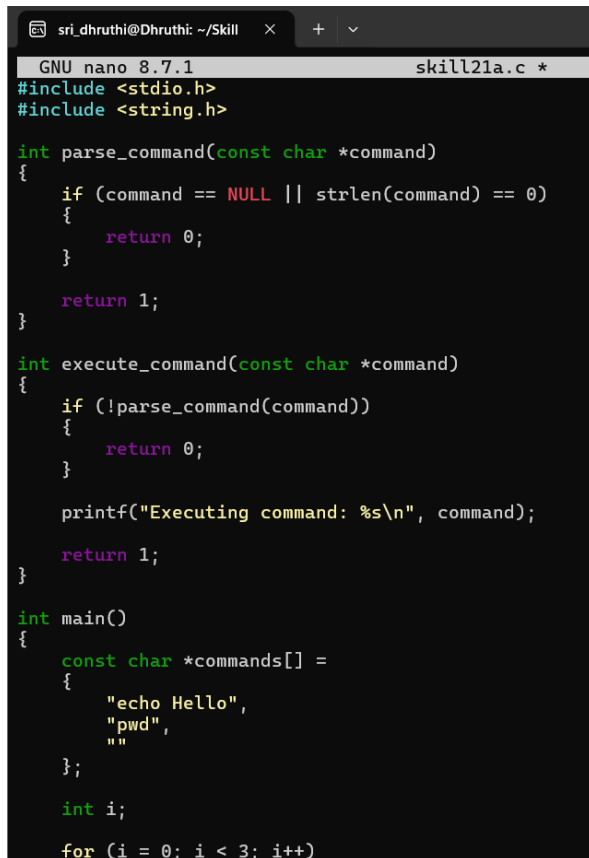


## Os skill 21

### 21(a) Connect Modules and Validate Interfaces

**What it does:** Connects simple shell modules, validates their interfaces, tests execution, and handles invalid input safely.



```
GNU nano 8.7.1 skill21a.c *
#include <stdio.h>
#include <string.h>

int parse_command(const char *command)
{
    if (command == NULL || strlen(command) == 0)
    {
        return 0;
    }

    return 1;
}

int execute_command(const char *command)
{
    if (!parse_command(command))
    {
        return 0;
    }

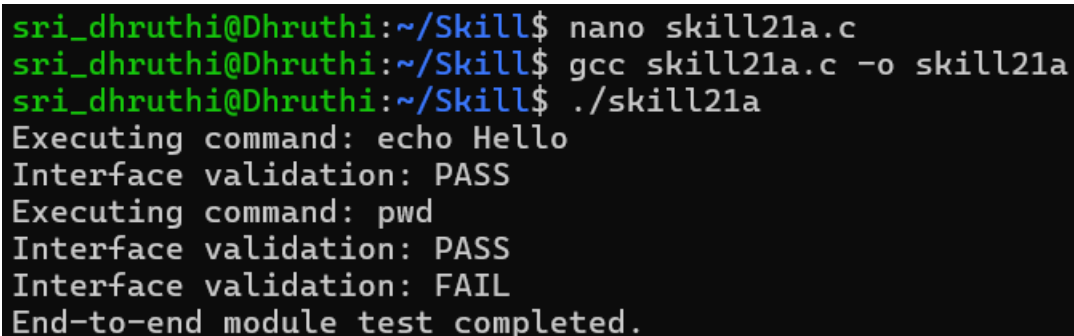
    printf("Executing command: %s\n", command);

    return 1;
}

int main()
{
    const char *commands[] =
    {
        "echo Hello",
        "pwd",
        ""
    };

    int i;

    for (i = 0; i < 3; i++)
```



```
sri_dhruthi@Dhruthi:~/Skill$ nano skill21a.c
sri_dhruthi@Dhruthi:~/Skill$ gcc skill21a.c -o skill21a
sri_dhruthi@Dhruthi:~/Skill$ ./skill21a
Executing command: echo Hello
Interface validation: PASS
Executing command: pwd
Interface validation: PASS
Interface validation: FAIL
End-to-end module test completed.
```

### 21(b) Runtime Error Handling

**What it does:** Detects runtime errors, displays an error message, and continues execution gracefully instead of crashing.

```
sri_dhruthi@Dhruthi: ~/Skill × + v
GNU nano 8.7.1 skill21b.c *
#include <stdio.h>
#include <stdlib.h>

int main()
{
    FILE *file;

    file = fopen("missing.txt", "r");

    if (file == NULL)
    {
        perror("Error opening missing.txt");

        printf("Recovery: File could not be opened.\n");
        printf("Program continues safely.\n");

        return 0;
    }

    printf("File opened successfully.\n");

    fclose(file);

    return 0;
}
```

```
End of the module test completed.
sri_dhruthi@Dhruthi:~/Skill$ nano skill21b.c
sri_dhruthi@Dhruthi:~/Skill$ gcc skill21b.c -o skill21b
sri_dhruthi@Dhruthi:~/Skill$ ./skill21b
Error opening missing.txt: No such file or directory
Recovery: File could not be opened.
Program continues safely.
sri_dhruthi@Dhruthi:~/Skill$ echo "Hello OS" > missing.txt
sri_dhruthi@Dhruthi:~/Skill$ ./skill21b
File opened successfully.
sri_dhruthi@Dhruthi:~/Skill$ |
```